

Gate E1 report — run_v1.21_r1 (Paper IV v1.21)

Verdict: **PASS (1 MINOR)**. Evidence: 20_EVIDENCE/E1/.

E1 — Claims revalidation (v1.21)

Basis: the auditor's diff v1.2 → v1.21 (EN and ES), the E1 claim map of run_v1.2_r1 (reused for unchanged statements), and the E6 automated parity results.

- **Statements, hypotheses, constants, tags and Lean blocks:** unchanged. Every theorem block (A, B, C, C', 3.1, 3.5, 5.0, 6.1, 6.1a, 6.2, 6.3, 6.3a, 6.4, 6.5, 6.6, D.1–D.3, F.3a/b, G.1) is identical. The display formulas are identical except for the declared subscripts N_{far} (formerly N_{lej} , N_{lejano}), e_{cross} , x^{clean} and E_{covered} , plus the defined abbreviation mis_c . These renamings are semantically neutral: each denotes the same object as before, and N_{far} was already the name used in (C.4) and (F.1). The run_v1.2_r1 claim map therefore stands.
- **Abstract / Theorem A remark:** now cites [5] and [15, Cor. 1.2] for the all-orders unrestricted bound. Accurate.
- **Front-matter status and §7 / A.1 status paragraphs:** they say the v1.2 external run was INCONCLUSIVE overall with formal reproduction PASS (607 modules, 19 targets, 50-module annex, 607 byte-identical objects), and that v1.21 needs its own review. Accurate per run_v1.2_r1.
- **New expository passages** (C.2, C.3, E.1): they introduce no new claim beyond the derivations reviewed in E2.
- **A.2 scope paragraph:** "the remaining tower iteration is a symbolic comparison with the recurrence in §6.3". §6.3 contains no recurrence, only T and h, so this cross-reference is inaccurate (part of A2-4 / X-05r).
- **NEW-01 (MINOR):** ordinary words have been set as code identifiers in headings, apparently by the monospacing pass:
 - EN §5.1 heading "Regularization on the original graph" and the A.2 table row "Regularization mass, degree, ...". In v1.2 both were plain words.
 - ES Lemma 3.2 heading "Nibble de rango acotado...", whereas the EN heading says "Bounded-rank nibble" in plain text.

The headings then read as if they referred to the Lean module `PaperIV.Regularization` or the library `Nibble`. It also creates an EN/ES heading inconsistency. Conversely, "bounded gain function from Model" lost the backticks it had on `Model` in v1.2 (§7.2), although it refers to the Lean namespace `Model` (OBSERVATION).

E1 verdict: PASS with one MINOR finding (NEW-01). No semantic change to any claim.